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Letter

Enzymatic Synthesis of Kynurenic Acid-Linked Dipeptides with C-Terminal Amide

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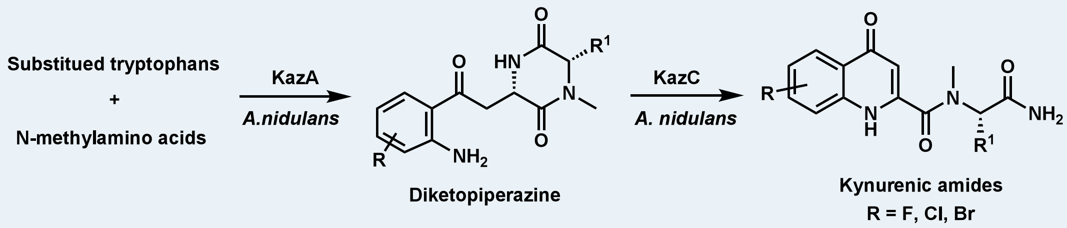

ABSTRACT: Kynurenic acid (KYNA) ester/amide derivatives were explored for improved biological potency. Herein, we report the discovery of a nonribosomal peptide synthase and P450 enzyme cascade capable of synthesizing KYNA-linked dipeptides in *Aspergillus nidulans*. *In vitro* assay together with isotope feeding experiments reveals an intriguing rearrangement from diketopiperazine (DKP) to dipeptide with a C-terminal amide. Feeding tryptophan derivatives allows the generation of various KYNA ester/amide derivatives. Our findings reveal the natural enzymatic machinery for KYNA amide biosynthesis, establishing a foundation for enzymatic access to KYNA amide derivatives.

KEYWORDS: *kynurenic acid amide derivatives, C-terminal amide, diketopiperazine cleavage, dipeptide, nucleophilic amination*

INTRODUCTION

Kynurenic acid (KYNA) is an endogenous excitatory amino acid receptor blocker and has demonstrated neuroprotective activity.[1] KYNA esters and amide derivatives have been investigated to enhance biological potency and selectivity.[2] However, these derivatives are usually synthesized via multistep reactions requiring harsh reagents (Figure 1),[3] as substituted KYNAs themselves are either commercially unavailable or prohibitively expensive. In particular, C-terminal amidation[4,5] represents a promising structural modification in KYNA conjugates, however, they have not been reported.

Biocatalysis offers a powerful approach for the efficient and sustainable production of complex molecules. Numerous structurally and functionally diverse protein families involved in amide bond synthesis have been characterized.[6] Among them, nonribosomal peptide synthetases (NRPSs),[7] N-acyltransferase,[8] aminoacylase[9] and lipase[10] are well investigated in small molecular amide bond formation. Whereas, neither NRPS mediated KYNA amide derivatives nor available transacylase that recruits KYNA has been identified to date. We thus try to identify enzyme candidates from natural product biosynthetic pathways. However, only a few natural 3-hydroxyl-KYNA amide derivatives have been isolated from a *Streptomyces* species,[11] and its unavailability hinders from further biosynthetic investigation. Therefore, we prioritized identifying more KYNA amide derivatives from microbial sources before pinpointing potential enzyme candidates.

We first employed genome mining, a powerful tool that uses biosynthetic pathway specific genes as a probe to identify structurally specific natural products, to explore KYNA amide derivatives.[12] In primary metabolism, KYNA is biosynthesized from L-tryptophan by the kynurenine pathway.[13] First, the tryptophan 2,3-dixoygenase (TDO) or indoleamine 2,3-dioxygenase (IDO), catalyzes oxidation of tryptophan to form N-formyl-L-kynurenine, followed by the removal of the formyl group by formamidase to give kynurenine, which is catalyzed by kynurenine amino transferase (KAT) to generate KYNA. We therefore use the rate- limiting IDO as a query to search for KYNA containing natural products; the result was refined by clustered NRPS or N-acyltransferase enzyme for potential amide bond formation. Finally, we identified a possible gene cluster, designated *kaz* containing six biosynthetic genes (Figure 2A, Table S3). These genes encode an NRPS (KazA), two methyltransferases (KazB and KazE), two $P450_s$ (KazC and KazF) and IDO (KazD). The NCBI genome database contains several clusters homologous to *kaz*, and each of them minimally contains four of the above genes (*kazA-D*) (Figure S1).

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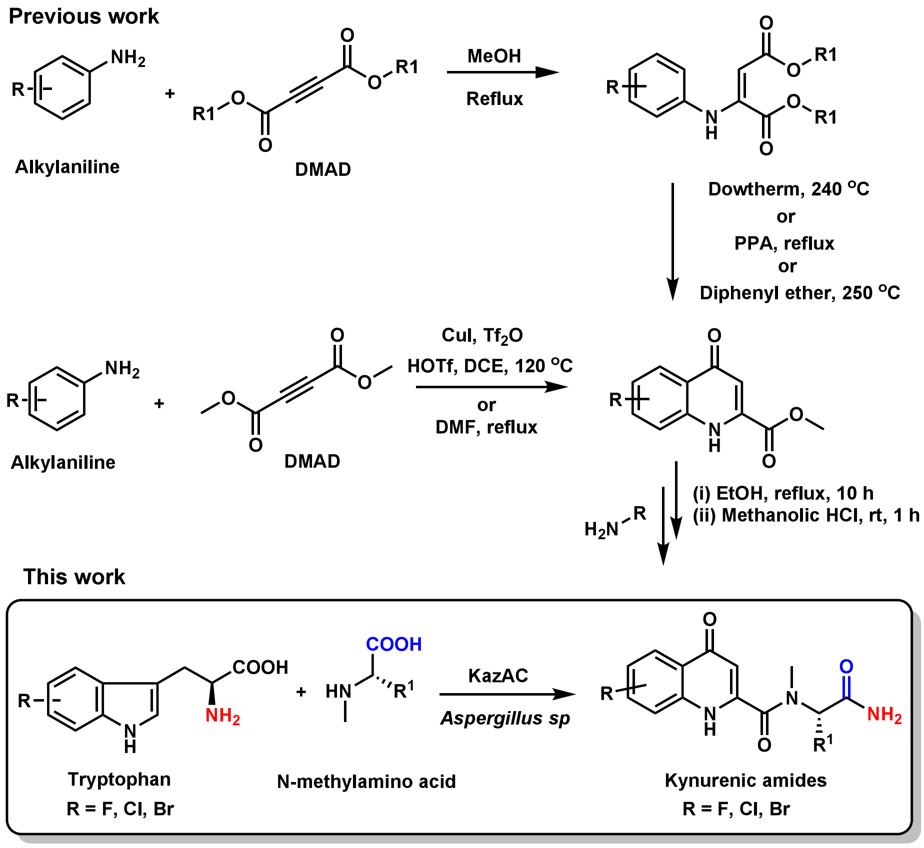

Figure 1. Approaches used to synthesize the kynurenic ester/amide derivative.

To identify the products assembled by the *kaz* cluster, we expressed the whole *kazA-F* in *Aspergillus nidulans* A1145.[14] Expectedly, two novel compounds, **1** and **2**, were produced in mutant strain expressing the whole *kaz* cluster (Figure 2B). Exclusion of IDO KazD from the coexpression strain results in decreased production of **1** and **2**, indicating that endogenous IDO of *A. nidulans* might complement the function of KazD (Figure S3). Scaled-up fermentation, followed by isolation and NMR characterization, confirmed that **1** is a dipeptide featuring a KYNA amide bond with a C-terminal amide (Table S8). While **2** is the deamidated analogue of **1** (Table S9). To the best of our knowledge, **1** and **2** represent the first KYNA linked peptides identified from fungi as well as the first KYNA conjugates with C-terminal amide to date. Endogenous amidases have been reported to catalyze amide bond hydrolysis;[15] we hypothesize that **2** is the hydrolysis product of **1** by endogenous amidases probably. Eight amidases from *A. nidulans* A1145 were then cloned and expressed, as expected, two of them can convert **1** to **2** (Table S4, Figure S4). This result supports the idea that **1** is the exact product of the *kaz* cluster.

With KYNA amide **1** in hand, we moved to investigate its biosynthetic machinery step by step. Coexpression of NRPS KazA and KazB yields diketopiperazine (DKP) compound **3** (Figure S3, Table S10).[16] Purified KazE (Figure S5) can methylate both kynurenine and **3** to the corresponding methylated products *N*-methyl kynurenine **4** (Figure S6) and **5** (Table S12), respectively. Feeding **4** (Table S11) into *kazAB* expressing strain results in the production of **5** (Figure S7), indicating both kynurenine and **4** can be recruited by KazA. Kinetic analysis of KazE toward kynurenine and **3** suggests that the k_{cat}/K_m is about 22 M^{-1} s^{-1} for kynurenine, and 156 M^{-1} s^{-1} for **3**, respectively, indicating the DKP formation is prior to methylation during *kaz* biosynthesis (Figure S8).

Coexpression of *kazABC* yielded KYNA amides **8** (Figure S3, Table S15) and **9** (Table S16), which are demethylated products of the corresponding **1** and **2**. Unexpectedly, we observed an intermediate **7** at day 3 (Figure S12), which almost completely disappeared by day 5-our standard time point for metabolite analysis. **7** (Table S14) is further characterized by NMR and HRMS to be the C7-hydroxyl product of **3**. Inspired by this finding, we reanalyzed the time course of the *kazABCE* coexpression strain, leading to the detection of **6** (Figure 2D) (Table S13). Consistently, KazC microsome prepared from *kazC* expression *A. nidulans* can convert DKPs **3** and **5** into the corresponding dipeptides **8** and **1** in Tris-HCl (pH 8.5) in the presence of NADPH and FAD, respectively (Figure 2C, Figure S9). Notably, putative intermediates **7** and **6** could not be detected even when the reaction time was shortened (Figure S10). To determine whether KazC is responsible only for the hydroxylation or also for the subsequent dipeptide formation, we tested its activity toward **6** and **7**. As a result, **6** (or **7**) rapidly convert to **1** (or **8**) even without KazC (Figure 2C, Figure S12). Further stability assays revealed that **6** (or **7**) is chemically labile under basic conditions but remains stable in aprotic solvents (Figure S14). The transformation of **6** to **1** is pH-dependent, the yield in PBS was ~12% at pH 6.0, and increased to ~18% at pH 7.0, and ~60% at pH 8.0 when incubated at 28 °C for 48 h (Figure S15). The neutral to slightly acidic intracellular environment of *Aspergillus* is consistent with the accumulation of **6** at an early stage. Notably, at pH 8.0, the conversion rate in NaOH-

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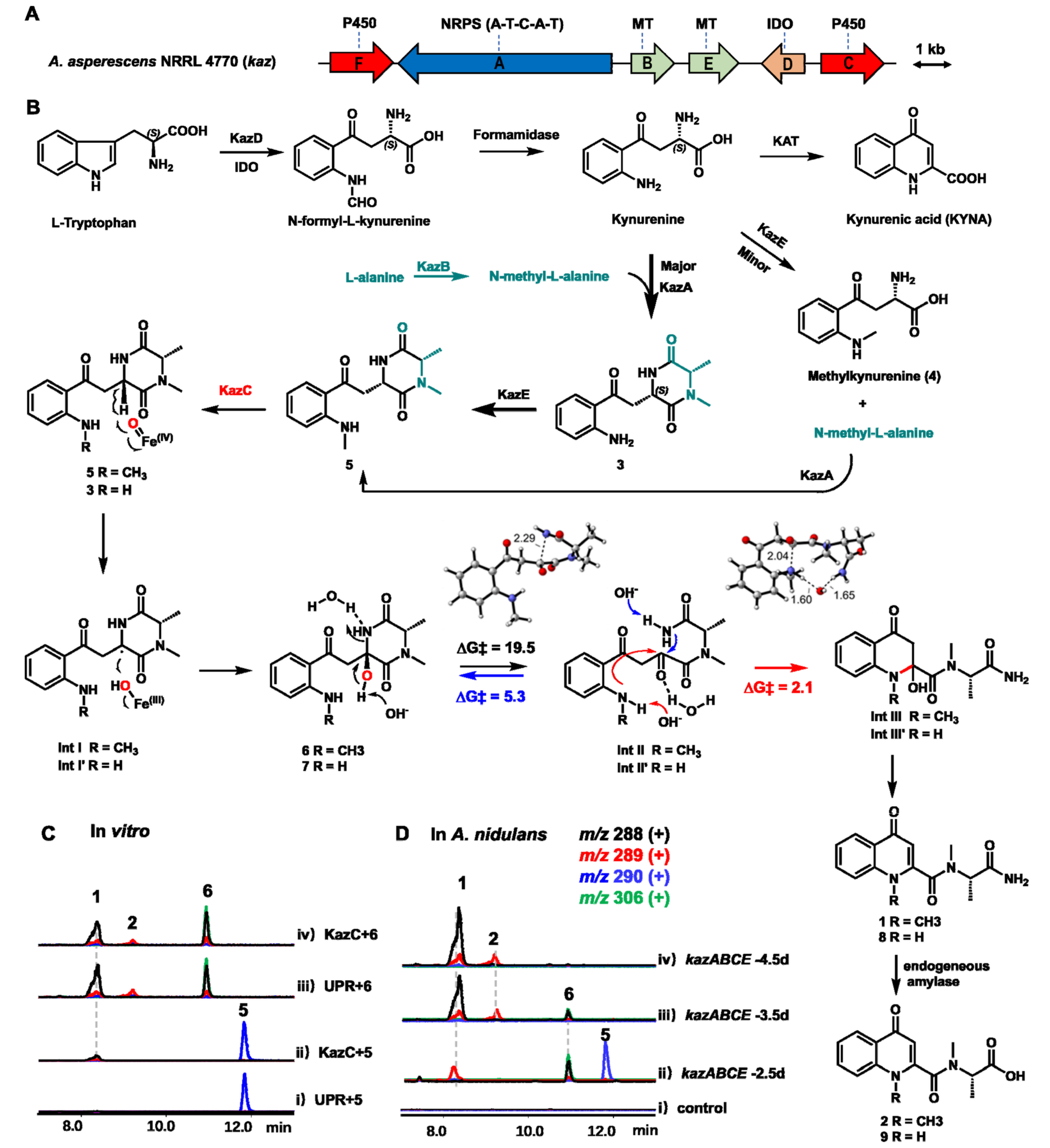

Figure 2. Biosynthetic pathway of compounds **1** and **2**. (A) The *kaz* cluster from *Aspergillus asperescens* NRRL 4770. (B) Proposed biosynthetic pathway to **1** and **2**; LC-MS analysis of transformation of **5** to **1** and **2** using KazC microsome (C) and *in vivo* heterologous expression (D). Extracted ion chromatograms (EICs) at m/z 288 contain signals from both **6** (MW 305), detected as the dehydrated ion $[M+H-H_2O]^+$, and **1** (MW 287), detected as $[M+H]^+$. UPR represents the cell lysis of the strain expressing empty vectors. DFT calculations at M06-2*X*/aug-cc-pVTZ/SMD(H_2O)//M06-2*X*/6-31+G(d,p)/SMD(H_2O). Free energies are in kcal mol^{-1} and bond lengths are in Å.

adjusted water (2.6×10^{-7} M^{-1} s^{-1}) was much lower than buffered solutions, with Tris-HCl (2.6×10^{-6} M^{-1} s^{-1}) consistently promoting faster conversion than PBS (1.5×10^{-6} M^{-1} s^{-1}) (Figure S15). Collectively, above results indicate that KazC catalyzes the hydroxylation of **5** to yield **6**, and **6** undergoes a pH-dependent, general acid−base catalysts-assisted DKP ring opening to afford **1**. Given the stability of the DKP core scaffold, to the best of our knowledge, our results reveal the first example of cleavage of the *sp3* C−N bond of DKP to the dipeptide with a C-terminal amide.

P450 KazC shares 45% amino acid sequence identity with monooxygenase AstF in sesquiterpene astellolide biosynthesis (Figure S17).[17] Further phylogenetic analysis of KazC suggests that it clusters with a DKP dehydrogenase EchP450,[18] whose

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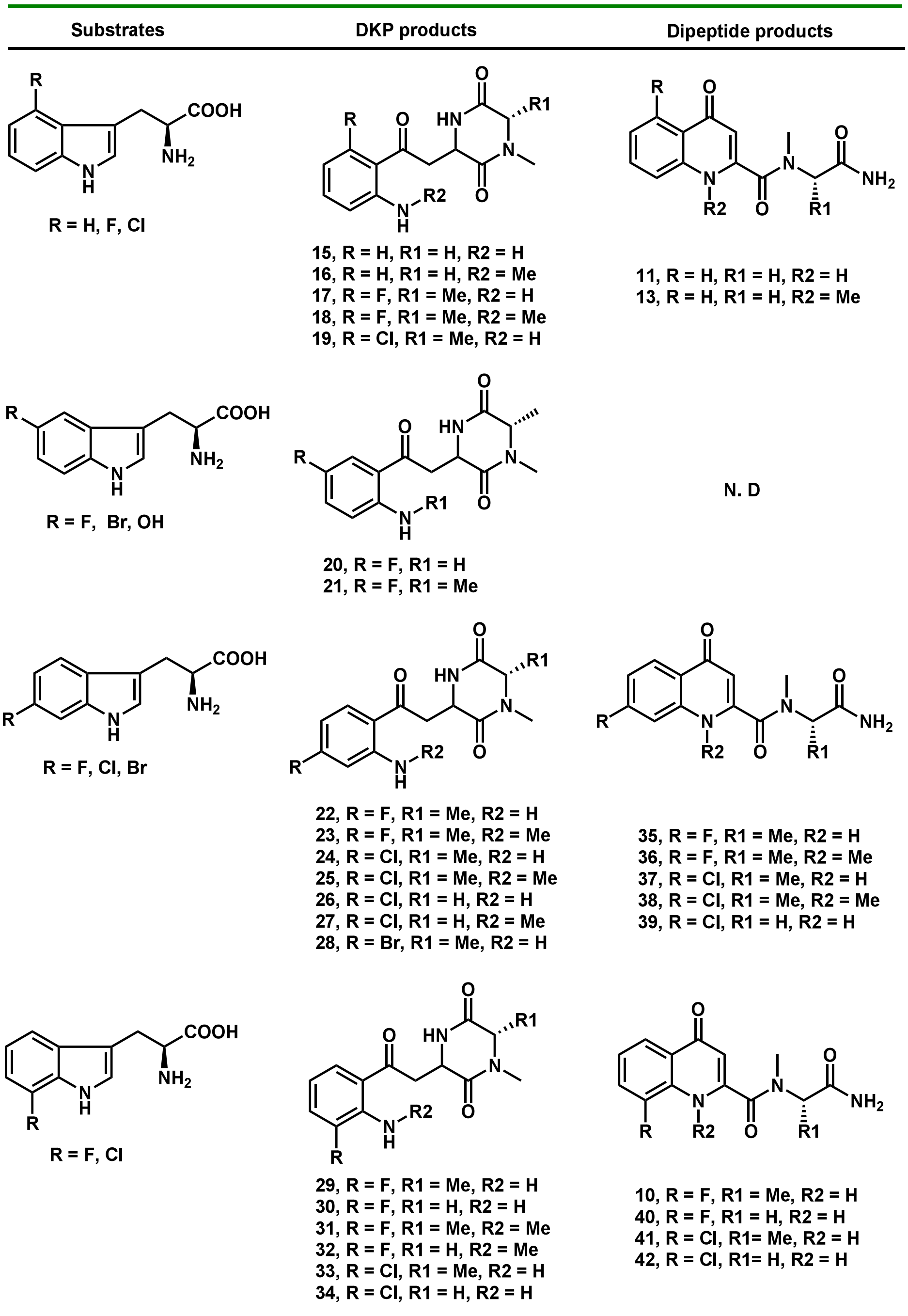

Figure 3. Substrate scope of *kaz*. Compounds **10**−**42** were generated by feeding substituted tryptophans and methylated amino acids.

function has only been confirmed *in vivo* (Figure S16). To reveal the reaction mechanism of DKP to dipeptide, we employed isotope labeled experiments. A mass shift of +1 Da was observed in **8** and **9** upon feeding with L-tryptophan-1-^{15}N, indicating the incorporation of the labeled pyridine nitrogen. Whereases L-tryptophan-^{15}N2 labeled assay results in a mass increase of +2 Da in **8**, and a mass increase of +1 Da in **9**, suggesting that the terminal NH_2 is from α-NH_2 of tryptophan (Figure S18). Based on above results, we propose the reaction starts from the homolysis of the C7−H of **3** (or **5**) by a ferryl-oxo specie to generate the C7 radical, OH rebound generated the hydroxylated DKP **7** (or **6**), identical to reactions catalyzed by known DKP hydroxylases such as OkaG[19] and FtmG.[18] In contrast to previously reported hydroxyl-DKP-type compounds—which either remain intact[19,20] or undergo hydroxyl-to-thiol substitutions[21] without cleavage of the DKP ring. **6** and **7** undergo further DKP scaffold disruption via the *sp*3 C−N bond cleavage, resulting in the formation of a terminal amide intermediate (**Int II**). Due to resonance delocalization with the adjacent carbonyl, the nucleophilicity of the amide nitrogen in **Int II** is significantly diminished, rendering the reverse cyclization to **6** and **7** less favorable. Consequently, the more nucleophilic *N*-methylaniline (or aniline) attacks the carbonyl intermediate, driving the pathway toward N17−C7 bond formation, followed by rearomatization that furnishes **1** and **8** (Figure 2B), resembling

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the transformations catalyzed by KAT enzymes in the primary pathway. Our hypothesis was further supported by density functional theory (DFT) calculations.[22] The C−N bond cleavage of **6** yields **Int II** (**TS-1**, $\Delta G‡$ = 19.5 kcal·mol^{-1}). The reverse reaction by terminal amide attack (**TS-1**, $\Delta G‡$ = 5.3 kcal·mol^{-1}), is 2.1 kcal·mol^{-1} higher in energy than nucleophilic addition by *N*-methylaniline (**TS-2**, $\Delta G‡$ = 3.2 kcal·mol^{-1}) (Figure 2B, Figure S19), indicating that formation of **1** is kinetically preferred. Consistently, **1** is 24.5 kcal·mol^{-1} lower in Gibbs free energy than **6**, rendering the transformation from **6** to **1** favored thermodynamically.

C-terminal amidation can improve pharmacokinetics by enhancing receptor binding affinity, and is critical in peptide hormones and neuropeptides maturation.[4] Various chemical and enzymatic methods—including electrochemical approaches[23] and recombinant protein cleavage[24] have been developed to synthesize C-terminal amides. Enzymatic amidation is primarily limited to two families of enzymes: a) amidotransferases,[25] which directly convert carboxylate groups into amides; b) peptidylglycine α-amidating monooxygenase (PAM),[5,26] which catalyzes α-hydroxylation followed by dealkylation of terminal glycine residues, is the main strategy in animals. Recently, enamine dealkylation has been reported for C-terminal amide formation.[27] We define a P450-initiated route to peptide C-terminal amidation derived from DKP-specific structural features, which is mechanistically analogous but distinct from the canonical Cu-dependent PAM system.

To assess the substrate promiscuity of the *kaz* pathway, we fed ten commercial tryptophan derivatives and seven *N*-methyl amino acids to strains expressing different gene combinations, respectively (Figure 3, Table S5, Figure S20). LC-MS analysis showed that eight of the ten substituted tryptophans and *N*-methyl glycine were accepted by the *kazAB* strain, which together leads to 20 new DKPs (Table S5, Figure 3). The remaining 5-hydroxy, and 5-bromotryptophan were not taken up by the cells (Figure S21). While only 11 of the 20 new DKPs were converted to their corresponding dipeptides by KazC, indicating KazC is the bottleneck enzyme to expand substituted KYNA amides.

Subsequent scaled-up fermentation and purification allowed structural characterization of six dipeptides including 7-fluoro-KYNA amides (**10**) (Table S17) and (**40**) (Table S24), 6-fluoro-KYNA amide (**35**) (Table S22), 6-chloro-KYNA amide (**38**) (Table S23), KYNA-Gly amides (**11**) (Table S18) and (**13**) (Table S20). With these KYNA amide derivatives in hand, we conducted a preliminary parallel artificial membrane permeability assay (PAMPA) to evaluate their blood−brain barrier (BBB) permeability.[28] Although none of these them exhibited improved BBB permeability (Table S7), C-terminal amide−containing peptides are usually associated with improved biological activity.[4] These KYNA conjugates with the C-terminal amide represent an unexplored scaffold with promising biological potential.

CONCLUSION

Collectively, we generated a series of novel KYNA amides with a C-terminal amide and elucidated their biosynthetic machinery. The enzymatic cleavage of DKPs to dipeptide is unprecedent, as it represents the reverse of peptide degradation, in which peptides are typically converted into DKPs under heating or in the presence of catalysts.[29] Our findings provide a valuable and versatile biocatalytic platform for the efficient synthesis of KYNA amide derivatives from readily available tryptophan derivatives. Further improvements in catalytic efficiency and substrate scope through enzyme engineering are anticipated to enhance the practical applicability of this platform.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.6c00080.

Experimental details, spectroscopic and computational data (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

KYNA, kynurenic acid; BBB, blood−brain barrier; NRPS, nonribosomal peptide synthetase; TDO, tryptophan 2,3-dixoygenase; IDO, indoleamine 2,3-dioxygenase; KAT, kynurenine amino transferase; DKP, diketopiperazine

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